Joonsuk Bae

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ALICE Run 3 charged-particle jets — end-to-end pipeline for ALICE's first Run 3 inclusive charged-particle jet cross-section measurement at $\sqrt{s} = 13.6$ TeV (preliminary; paper in preparation; EPS-HEP 2025 jet working group merge talk on behalf of ALICE). Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.

Education

Ph.D. Candidate, Physics — Sungkyunkwan University   2022.03 – expected 2027.02

Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE.* Defense expected 2026.12; degree conferral 2027.02.

B.S., Physics — Sungkyunkwan University   2021.02

Appointments and Collaboration Membership

ALICE Collaboration, CERN, Geneva	2022.10 – present
ePIC Collaboration, Brookhaven National Laboratory	2024.02 – present
LAMPS Collaboration, RAON, Korea	2022.03 – 2024.04

Research Experience

Main analyzer, ALICE Run 3 charged-particle jets (PWG-JE) — Sungkyunkwan University   2023.04 – present

- **First ALICE Run 3 inclusive charged-particle jet cross section** at $\sqrt{s} = 13.6$ TeV — responsible for the full chain: O^{2}Physics task development, QA on continuous-readout data, Bayesian/SVD unfolding, and the systematic-uncertainty program. ALICE Preliminary; paper in preparation. Varied $R = 0.2$–0.6 to separate collinear from large-angle radiation.
- **Tracking-efficiency systematics for Hard Probes 2026** (ALICE jet working group) — derived the pp charged-particle tracking-efficiency uncertainty at $\sqrt{s} = 13.6$ and 5.36 TeV (p_{T}-binned, 2022–2024 periods), combining the track-selection component with the ITS–TPC matching component.
- **Track momentum resolution** — studied data–MC resolution mismatch by covariance propagation, MC residuals, and V0 mass resolution, and derived the p_{T}-dependent smearing profile used in Run 3 charged-jet systematics.
- **Cross-section normalization** — implemented the Run 3 charged-jet normalization from the ALICE TVX visible cross section under continuous readout, quantifying the bunch-crossing and track-matching inefficiencies introduced by time-frame-based asynchronous reconstruction and deriving the corresponding signal-loss corrections. Now becoming the standard normalization for Run 3 charged-jet analyses in the group.
- **Event-generator production for model comparison** — produced and managed the comparison sets behind the Run 3 jet-radius measurement, spanning NLO matching (POWHEG-BOX), leading-order tune variations (PYTHIA 8: Monash, colour reconnection, rope hadronization, string shoving), and an independent parton-shower and hadronization model (HERWIG 7), with provenance tracking across configurations.

Detector physicist, ePIC Barrel Imaging Calorimeter (BIC) — Korea-BIC group   2024.02 – present

- **Pb/scintillating-fiber prototype test beams** — lead analyst for the CERN PS T10 (2025.07) campaign: energy response, resolution, and linearity. On-site DAQ shifter with equalization and prompt analysis at KEK AF–AR (2025.03), and at the first PS T10 campaign (2024.08), published as a co-author.
- **Module QA and assembly** — ran the PMT-module QA for the prototype modules before the test beams (cosmic-ray and source characterization, timing-coincidence checks, optical-coupling stability, per-module acceptance criteria); Pb/scintillating-fiber bundle alignment, polishing, and PMT/SiPM coupling.

- Energy- and η-dependent sampling-fraction studies in JANA2 / EIC-recon for the Korea-BIC simulation effort; Geant4 prototype modeling.

Detector R&D, LAMPS Forward Tracker — RAON, Korea 2022.03 – 2024.04

- Early-stage design and Geant4 simulation of an MPPC–scintillator-plane forward tracking detector; sensor surveys, specifications, and tracking-concept feasibility studies.

Selected Publications

INSPIRE: inspirehep.net/authors/2117232 · ORCID: 0009-0008-4806-8019. ALICE Collaboration co-author since 2022.10; ECCE/ePIC and LAMPS collaboration papers are included in the INSPIRE record.

- ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* (submission expected 2026). Thesis paper.
  [main analyzer]
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae** *et al.*, "Beam test of a Pb/SciFi prototype for the Barrel Imaging Calorimeter at the Electron-Ion Collider" — CERN PS T10 (2024.08) campaign; submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482); `arXiv:2604.22647`.
  [co-author; module-construction support and PMT/module performance characterization]
- Y. Hong, J. Bok, G. An, **J. Bae** *et al.*, "Test-beam performance of the AstroPix silicon sensor for imaging calorimetry," submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00572); `arXiv:2605.07681`.
  [co-author; beam-test DQM shifts and on-site prompt analysis]
- S. Y. Jang, G. An, **J. Bae** *et al.*, "Performance of dual-readout calorimeters for various absorbers," *Nucl. Instrum. Meth. A* (2026), DOI `10.1016/j.nima.2026.171598`.
  [co-author; beam-test participation and DAQ shifts]
- G. Cho, S. Kim, **J. Bae** *et al.*, "Beam tests of the copper-based dual-readout calorimeter to measure electromagnetic performance for future e^+e^- colliders," *J. Subatom. Part. Cosmol.* **3** (2025) 100021.
  [co-author; module construction and beam-test data taking]
- Y. J. Kim, D. S. Ahn, J. K. Ahn *et al.* (LAMPS), "Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON," *J. Korean Phys. Soc.* **87** (2025) 670–682.
  [co-author; Time-of-Flight detector maintenance and on-site checks]
- Y. Bae *et al.* (LAMPS), "Performance test of prototype LAMPS Start Counter using proton beams at 100 MeV," *Nucl. Instrum. Meth. A* (2025), DOI `10.1016/j.nima.2025.170827`.
  [co-author; Start-Counter prototype testing]
- Y. Cheon *et al.* (LAMPS), "Characteristics of the prototype LAMPS TPC for low-energy nuclear collisions at RAON," *Nucl. Instrum. Meth. A* **1066** (2024) 169610.
  [co-author; TPC prototype characterization]
- H. Kim *et al.* (LAMPS), "Performance of the prototype beam drift chamber for LAMPS at RAON with proton and ^{12}C beams," *JINST* **19** (2024) P12008; `arXiv:2412.08662`.
  [co-author; forward-tracking R&D and prototype performance studies]

Invited and Contributed Talks

Invited Talks (external workshops, seminars)

- **France–Korea Particle Physics Network (FKPPN) Workshop**, Inha University — *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* — 2025.12
- **1st Heavy-Ion Meeting (HIM 2025)**, APCTP — *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* — 2025.05
- **FKPPN Workshop**, Inha University — *Charged-jet studies in ALICE Run 3* — 2023.11

Collaboration-Meeting and National-Workshop Talks (internal)

- **Korea-BIC 2025 Summer Workshop** — *Data analysis of the recent beam test at CERN* — 2025.09
- **KoALICE National Workshop 2024** — *Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions* — 2025.01
- **ALICE Physics Week 2024** — *Charged-particle jet production in Run 3* — 2024.12
- **KoALICE National Workshop 2023** — *Charged-jet O^2 MC and data QA* — 2024.01
- **KoALICE National Workshop 2022** — *Dijet studies with ALICE at the LHC* — 2023.01

Contributed Talks and Posters — International Conferences

- **Hard Probes 2026**, Nashville — *Jet-like dihadron correlations in pO, OO, and Ne–Ne collisions with ALICE [talk]* — 2026.06
- **Hard Probes 2026**, Nashville — *Characterizing inclusive charged-particle jet production and fragmentation in pp collisions [poster]* — 2026.06
- **EPS-HEP 2025**, Marseille — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE [on behalf of the ALICE Collaboration; jet working group merge talk]* — 2025.07
- **INPC 2025**, Daejeon — *Charged-particle jet and jet quenching measurement in pp collisions at $\sqrt{s} = 13.6$ TeV during LHC Run 3 with ALICE [talk]* — 2025.05
- **Quark Matter 2025**, Frankfurt — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE [poster]* — 2025.04
- **Hard Probes 2024**, Nagasaki — *First measurements of charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE [poster]* — 2024.09
- **Quark Matter 2023**, Houston — *Charged-particle multiplicity distribution in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE [poster]* — 2023.09
- **ATHIC 2023** (9th Asian Triangle Heavy-Ion Conf.), Incheon — *Dijet studies at the LHC [poster]* — 2023.04

Contributed Talks and Posters — Korean Physical Society

- **KPS Spring 2026** *[talk]* — *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2026.04
- **KPS Fall 2025** *[talk]* — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.10
- **KPS Spring 2025** *[talk]* — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.04
- **KPS Spring 2025** *[poster]* — *Prototype calorimeter QC for the Barrel Imaging Calorimeter at the EIC* — 2025.04
- **KPS Fall 2024** *[talk]* — *First jet measurement in ALICE Run 3: charged-particle jet production* — 2024.10
- **KPS Spring 2024** *[talk]* — *Charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV* — 2024.04
- **KPS Fall 2023** *[talk]* — *First look at charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE Run 3 data* — 2023.10
- **KPS Spring 2023** *[poster]* — *Dijet studies at the LHC* — 2023.04
- **KPS Fall 2022** *[poster]* — *Dijet studies with ALICE at the LHC* — 2022.10

Service and Professional Activities

- **ALICE Run 3 O+O charged-jet measurement** — Paper Committee member (one of the principal authors); co-author on the Analysis Note; derived the cross-section normalization factor and its QA; co-mentor of the MSc main analyzer.
- **ePIC Korea-BIC beam-test analysis framework** — developed and shared the analysis framework across the Korean BIC institutes (SKKU, PNU, KNU).
- **ALICE ITS2 alignment monitoring** (Run 3) — ALICE service task: period-by-period tracking-performance diagnostics from compressed-time-frame (CTF) data.
- **KoALICE national activities** — regular participation since 2022, including invited collaboration-meeting talks.

Awards and Fellowships

- **BK21 FOUR Graduate Scholarship** — awarded on research performance by the Sungkyunkwan University Physics BK21 sponsored unit (Korean Ministry of Education), 2022 – present.

Teaching and Mentoring

Student mentoring (non-supervisory) — Sungkyunkwan University 2024 – present

- *Wooseok Ham* (M.Sc., main analyzer of the ALICE Run 3 O+O charged-jet measurement) — mentored on Run 3 jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; co-author on the corresponding ALICE Analysis Note.
- *Harim Seong* (undergraduate) — ePIC BIC detector-performance studies using JANA2-based reconstruction; energy response and sampling-fraction extraction.

KoALICE O^2 Tutorial — student co-organizer and instructor — Sungkyunkwan University 2024.01

- Co-organized (with H. Lee) the two-day Run 3 analysis-framework tutorial for the Korean ALICE community, ∼22 participants across Korean institutes; instructor on *Running O^{2}Physics with Hyperloop.*

Teaching Assistant, SKKU Physics (2022 – 2024) — *Nuclear Reactions* (graduate), *Quantum Mechanics II*, *General Physics I and Laboratory I.*

Technical Skills

Languages. English (professional working proficiency), Korean (native).
Programming. C++, Python, ROOT, LaTeX, Bash, Git.
HEP frameworks. ALICE O^{2}Physics, Geant4, JANA2 / EIC-recon, FastJet.
Software contributions. ALICE O^{2}Physics — C++ analysis tasks on the framework's columnar data model.
Event generators. PYTHIA 8, POWHEG-BOX, HERWIG 7.
Analysis methods. jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.
Detector hardware. PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.

References

Available on request.